
BENJAMIN DOURTHE, PhD

AI Strategist & Machine Learning Leader | Digital Health & MedTech | Team & Program Builder

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SUMMARY

AI and data science leader with 10+ years across digital health, MedTech R&D, and regulated environments. Built and scaled machine learning, computer vision, and signal-processing systems from prototype to product, including first-in-class wearable, implantable and imaging solutions. Known for fast, robust iteration, strong engineering discipline (automation, testing, reproducibility), and cross-functional leadership across software, engineering, clinical, and research teams.

CORE COMPETENCIES

Digital health & MedTech AI: biomedical sensing, medical imaging, digital biomarkers, wearables, implantables.

Agentic AI: platform rollout, prompt engineering, governance, security/compliance reviews.

Machine learning & deep learning: classification/regression, CNNs/RNNs, generative models, segmentation.

Computer vision & imaging: reconstruction, super-resolution (SRGAN), 2D/3D pipelines, quality optimization.

Time-series & signal processing: feature engineering, denoising, model robustness, generalization.

Production-grade Python: data pipelines, experiment tracking, code quality, automation, Git.

Leadership: hiring/interviewing, mentoring, roadmap delivery, stakeholder communication.

TECHNICAL SKILLS

Programming Languages: Python (primary), MATLAB, R, SQL, C, C++, Java, JavaScript, HTML/CSS.

Machine Learning & Deep Learning: scikit-learn, PyTorch, TensorFlow, hyperparameter optimization, model evaluation/validation, robustness testing, explainability.

Generative AI & Agentic Systems: Prompt and context engineering (advanced system prompts, reusable prompt frameworks), RAG design/build, org/team/project-specific skills; tools: Claude Code, OpenAI Codex, Gemini Enterprise, Cursor, Windsurf, GitHub Copilot, Deep Research.

Data, Signals & Imaging: NumPy, Pandas, SciPy, OpenCV, time-series analysis, feature engineering, multimodal data fusion, clinical-grade analytics.

DevOps, CI/CD & Engineering Practices: Git, testing, automation, documentation, reproducible research; CI/CD; project architecture strategies; data management for security/privacy/compliance; data & AI policy development.

Backend/Frontend & Full-Stack Foundations: APIs, backend and frontend frameworks, internal tooling and workflow automation for ops-heavy teams.

Domains: Digital health, MedTech, medical imaging, biomechanics, cardiac/pulmonary monitoring, wearables and sensor analytics; startup and early-stage product development.

PROFESSIONAL EXPERIENCE

STAFF MACHINE LEARNING ENGINEER, Shifamed / Adona Medical - Los Gatos, CA (2024 - Present)

MedTech company developing implantable cardiac monitoring systems.

- Lead enterprise rollout of Generative and Agentic AI platforms (strategy, enablement, and governance), including security/compliance reviews and internal controls to protect sensitive data and meet regulatory requirements.
- Design and deliver cross-functional Generative and Agentic AI training, handbooks, and office hours; internal survey measured ~20% average productivity lift across engineering and non-engineering teams, increasing throughput, creativity, and work quality while driving meaningful cost savings.
- Develop automated and agentic data-processing and analytics pipelines, as well as custom project- and team-specific agentic applications across Software, EE, and R&D to accelerate iteration, improve robustness, and provide near real-time operational insights.
- Engineer and validate signal processing and machine learning algorithms for complex cardiac pressure data to support next-generation clinical tools and data infrastructure.

- Founded and leading the AI & Data Science operating model across algorithm development/review, data standardization & governance, AI tool enablement/optimization, and custom agentic tools tailored to internal workflows.
- Built and maintained an internal agentic AI engineering toolkit with 277 production-ready prompt and context engineering templates across 7 languages, enabling consistent, high-quality AI-assisted development across teams.
- Authored compliance-oriented templates spanning SOC 2, ISO 27001, GDPR/CCPA, NIST AI RMF and agent governance patterns (lifecycle/risk/security/observability) to support regulated development practices.

SENIOR MACHINE LEARNING ENGINEER, Nuralogix Corporation - Toronto, Canada (2022 - 2023)

Digital health company using computer vision for video-based physiological monitoring.

- Led end-to-end design and optimization of signal-processing + ML pipelines, improving video data quality by 25% and biometric prediction performance by 12% (internal benchmark).
- Engineered 100+ high-value features for high-dimensional data representation, enabling stronger predictive performance and faster hypothesis testing.
- Introduced robustness-focused evaluation and model generalization strategies for multi-modal health data.

CHIEF DATA OFFICER, Tidal Medical - Toronto, Canada (2020 - 2022)

Startup developing first-in-class non-invasive wearable for real-time cardiopulmonary monitoring.

- Defined the data/AI architecture for a first-in-class non-invasive wearable for real-time pulmonary artery pressure monitoring, achieving 97% accuracy and enabling heart-failure management use cases.
- Pioneered deep learning approaches for medical imaging enhancement, including SRGAN-based super-resolution to improve diagnostic image quality.
- Improved software/algorithm efficiency by ~90% via redesign of signal-processing and computer-vision pipelines, enabling faster prototyping and more robust deployments.
- Built and led multidisciplinary teams; drove IP strategy and execution (1 PCT + 4 US provisional filings) and secured >\$2M in non-dilutive funding across national programs and grants.
- Established partnerships across academia and industry (e.g., Stanford, University of Toronto/Waterloo, Canadian Space Agency, Sciospec), accelerating translation from concept to prototype and product.

MACHINE LEARNING ENGINEER (Postdoctoral), University of British Columbia - Vancouver, Canada (2020 - 2022)

- Designed and deployed deep learning models for medical image segmentation (e.g., multi-scale architectures) improving accuracy and workflow efficiency for clinical research teams.
- Collaborated with corporate partners (Medtronic) and clinical stakeholders to deliver solutions suitable for real-world medical environments.
- Published peer-reviewed work in Spine (2022) on automated segmentation of spinal muscles from upright open MRI.

COMPUTER VISION ENGINEER (Postdoctoral), Sunnybrook Research Institute - Toronto, Canada (2018 - 2020)

- Developed marker-less human motion analysis using RNN/LSTM and computer-vision pipelines, improving tracking accuracy by 35% for non-invasive monitoring and rehabilitation applications.
- Integrated multiple sensing modalities to advance real-time biomechanical data analysis in clinical settings.

DATA SCIENTIST (Postdoctoral), University of Calgary - Calgary, Canada (2017 - 2018)

- Delivered data analytics and experimentation platforms bridging academia and industry (Adidas, Dr. Scholl's, CCM Hockey, Orpyx) for human performance and product development.
- Designed multi-sensor data collection and analysis strategies supporting multi-modal insights and decision-making.

RESEARCH ENGINEER (PhD), KU Leuven - Leuven, Belgium (2014 - 2017)

- Developed patient-specific computational models combining medical imaging and biomechanical simulation to predict degenerative musculoskeletal conditions (precision-medicine applications).
- Built proprietary image-processing algorithms to extract morphological features and map mechanical stress distributions, enabling risk stratification and clinically relevant visualization.
- Led high-fidelity data collection across cohorts; created novel 3D mapping workflows from CT imaging.

EDUCATION

- **PhD, Biomedical Engineering** - KU Leuven, Belgium (2014 - 2017)
Thesis: Computational modeling of biological systems; developed patient-specific predictive models combining imaging and biomechanical simulation.
- **MSc, Science, Technology and Health** - University of Lorraine, France (2012 - 2013)

- **MSc, Mechanical Engineering** - National Engineering School of Metz (ENIM), France (2008 - 2013)
- *Visiting Scholar* - University of British Columbia, Canada (Fall 2016); University of Southern California, USA (Fall 2015)

HONORS & AWARDS

- Secured 3rd prize at the Dreamscape hackathon organized by AGI House, sponsored by Coatue and Mantis (Oct 2023)
- Toronto Health Stream Graduate, Creative Destruction Lab (CDL) (Jul 2022)
- Deep Space Healthcare Challenge, Canadian Space Agency (Jun 2022) (\$30,000)
- Research and Development Award, National Research Council of Canada IRAP (Apr 2022) (\$250,000)
- INOVAIT Strategic Innovation Fund, Sunnybrook Research Institute (Mar 2022) (\$125,000)
- Novel applications of AI and image reconstruction for respiratory and cardiac monitoring (VIP Program), ENCQOR (Dec 2021) (\$150,000)
- Demonstration Program, ENCQOR 5G (Nov 2021) (\$50,000)
- Project Grant, Canadian Institutes of Health Research (Sep 2021) (\$795,600)
- Collaborative Research and Development Grant, NSERC (Jan 2018) (\$100,920)
- European Research Foundation Conference Grant, Research Foundation Flanders - FWO (Jan 2017) (EUR 1,300)
- European Research Foundation Visiting Scholar Grant, Research Foundation Flanders - FWO (Jan 2016) (EUR 5,000)
- Conference Grant, Academische Stichting Leuven (Jan 2015) (EUR 500)
- European Research Foundation Visiting Scholar Grant, YouReCa Junior Mobility Programme (Jan 2015) (EUR 4,000)
- Ouvertud Scholarship, Region Lorraine (Jan 2011) (EUR 800)
- Government Scholarship, French Ministry of Higher Education and Research (2009-2013) (EUR 19,500)
- Corporate Scholarships, Group IRP Auto (2009-2013) (EUR 4,000)

PATENTS & PATENT APPLICATIONS

- System and method for respiratory monitoring - US Provisional 63/213,194 (2021)
- Method and system for electrical impedance tomography - PCT CA2021050649 (2021)
- Increasing effective sampling rate of EIT recordings to improve accuracy of pulse transit time measurements - US Provisional 63/303,963 (2022)
- Wearable radar sensing system - US Provisional 63/303,882 (2022)

PUBLICATIONS, INDUSTRY REPORTS & INDEPENDENT RESEARCH

- Dourthe, B. (2025). The Future of Intelligence: The Embodiment Gap and the Path to World Models. Zenodo. <https://doi.org/10.5281/zenodo.18177869>
- Dourthe B, Ashjaee N, Fels S, Oxland TR. Automated detection of vertebral body centroids from CT-based reconstructed X-ray images using deep convolutional neural network. Unpublished (2023).
- Dourthe B, Stowe S, Sawyer A. Pulmonary artery pressure estimation using electrical impedance tomography. Tidal Medical (2022).
- Toma J, Sawyer A, Pelletier S, Pallopson C, Dourthe B, Stowe S, Adler A, Grychtol B. Upsampling technique for generating electrical impedance tomography frames with increased temporal resolution. Tidal Medical (2022).
- Dourthe B, Shaikh N, Pai SA, Fels S, Brown SHN, Wilson DR, Street J, Oxland TR. Automated segmentation of spinal muscles from upright open MRI using a multiscale pyramid 2D convolutional neural network. Spine. 47(16) (2022) 1179-1186.
- Dourthe B, Pelletier S. Deep learning super resolution approaches for medical imaging enhancement. Tidal Medical (2021).
- Toma J, Adler A, Sung A, Dourthe B, Sawyer A, Pelletier S, Ardeshiri R, Pallopson C. Method and system for electrical impedance tomography. Tidal Medical (2021).
- Dourthe B, Burns D, Wasserstein D, Whyne C, Hardisty M. Using recurrent neural network for time series regression: towards the enhancement of computer vision-based 3D motion sensing technologies for injury risk assessment. Unpublished (2020).
- Dourthe B, Brenkel M, Ziegler T, Wu LY, Wasserstein D, Zabjek K, Beach T, Saumur T, Hirsch SM, Whyne C, Hardisty M. Comparing marker-based 3D motion capture to computer vision-based motion sensing technologies to assess biomechanical variables associated with anterior cruciate ligament injury risk. Unpublished (2020).
- Dourthe B, Osterloh J, von Tscharnner V, Nigg S, Nigg B. Impact of self-selected customized orthotics on lower limbs biomechanics. Prosthetics and Orthotics, Open-Access Peer-Reviewed Chapter (2020).
- Dourthe B, Nigg S, Nigg BM. Highlights of the Fourteenth Footwear Biomechanics Symposium. Dr. Scholl's, Biomechanigg Sport & Health Research (2019).

- Dourthe B, Osterloh J, von Tscharner V, Nigg S, Nigg B. Impact of self-selected customized orthotics on lower limbs biomechanics and fatigue. *Footwear Sci.* 11 (2019) S22–S23.
- Manz S, Dourthe B, Nigg S, Nigg B. The influence of preference grouping methods on characteristics of recreational runners. *Footwear Sci.* 11 (2019) S40–S42.
- Dourthe B, Nickmanesh R, Wilson DR, D'Agostino P, Patwa AN, Grinstaff MW, Snyder BD, Vereecke E. Assessment of healthy trapeziometacarpal cartilage properties using indentation testing and contrast-enhanced computed tomography. *Clinical Biomechanics.* 61 (2019) 181–189.
- Osterloh J, Dourthe B, Nigg S, Nigg BM. Measuring fatigue-induced changes in walking kinematics: impact of different insole configurations. Master Thesis Chapter, Unpublished (2019).
- Dourthe B, Osterloh J, von Tscharner V, Nigg S, Nigg B. Quantifying the effects of different insole configurations on fatigue and comfort - Phase 3: Results following a sub-hour inclined power walking intervention. Dr. Scholl's, Biomechanigg Sport & Health Research (2018).
- Dourthe B, Osterloh J, von Tscharner V, Nigg S, Nigg BM. Quantifying the effects of different insole configurations on fatigue and comfort - Phase 2: Results following a 45-minute power walking intervention. Dr. Scholl's, Biomechanigg Sport & Health Research (2018).
- Meixner A, Dourthe B, Kerkhof F, Nigg S, Nigg BM. Comparison of a newly introduced pressure insole to existing foot pressure measurement technologies. Orpyx Medical Technologies Inc., Biomechanigg Sport & Health Research (2018).
- Meixner A, von Tscharner V, Dourthe B, Nigg S, Nigg BM. Advanced analysis of the EMG data of the vastii muscles during treadmill running. Biomechanigg Sport & Health Research (2018).
- Dourthe B, Nigg S, Meixner A, Osterloh J, Hoitz F, Nigg BM. Quantifying the effects of different insole configurations on fatigue and comfort - Phase 1: Results following a single working shift intervention. Dr. Scholl's, Biomechanigg Sport & Health Research (2018).
- Dourthe B, Nigg S, Lam C, Fletcher JR, Martinez A, Omu O, Nigg BM. Effect of compression apparel on variables associated with performance, posture, comfort and fatigue. Per4ma Sports Technology LTD, Biomechanigg Sport & Health Research (2018).
- D'Agostino P, Dourthe B, Kerkhof F, Stockmans F, Vereecke E. Impact of osteoarthritis and total joint arthroplasty on the kinematics of the trapeziometacarpal joint: a pilot study. *Journal of Hand Surgery.* 43 (2018) 382.e1–382.e10.
- D'Agostino P, Dourthe B, Kerkhof F, van Lenthe GH, Stockmans F, Vereecke E. In vivo biomechanical behavior of the trapeziometacarpal joint in healthy and osteoarthritic subjects. *Clinical Biomechanics.* 49 (2017) 119–127.
- D'Agostino P, Dourthe B, Kerkhof F, Stockmans F, Vereecke E. In vivo kinematics of the thumb during flexion and adduction motion: Evidence for a screw-home mechanism. *Journal of Orthopaedic Research.* 35 (2017) 1556–1564.
- Dourthe B, Valero-Cuevas FJ, D'Agostino P, Kerkhof F, Vereecke E. Towards a forward simulation of transverse loading of the trapeziometacarpal joint. Dissertation Chapter, Unpublished (2016).
- Stockmans F, Dourthe B, Vanovermeire O, Deillie M, Vanhaecke J, D'Agostino P, Vereecke E, Kerkhof F. Influence de la reconstruction du ligament radio-dorsal sur la stabilité de l'articulation trapeziometacarpien. *Hand Surgery & Rehabilitation.* 35 (2016) 453–454.
- Kerkhof F, Brugman E, D'Agostino P, Dourthe B, van Lenthe GH, Stockmans F, Jonkers I, Vereecke E. Quantifying thumb opposition kinematics using dynamic computed tomography. *Journal of Biomechanics.* 49 (2016) 1994–1999.
- Dourthe B, D'Agostino P, Stockmans F, Kerkhof F, Vereecke E. In vivo contact biomechanics in the trapeziometacarpal joint using finite deformation biphasic theory and mathematical modelling. *Medical Engineering & Physics.* 38 (2016) 108–114.

CONFERENCE PRESENTATIONS & INVITED TALKS

- Using computer vision sensing technologies for anterior cruciate ligament injury screening, 18th annual Imaging Network Ontario Symposium, Toronto, ON, Canada (poster presentation, Mar 2020).
- Impact of self-selected customized orthotics on lower limbs biomechanics and fatigue, 14th biennial Footwear Biomechanics Symposium, Kananaskis, AB, Canada (oral presentation, Jul 2019).
- The influence of preference grouping methods on characteristics of recreational runners, 14th biennial Footwear Biomechanics Symposium, Kananaskis, AB, Canada (oral presentation, Jul 2019).
- Correlation between thumb cartilage properties and medical imaging, 8th World Congress of Biomechanics, Dublin, Ireland (poster presentation, Jul 2018).
- First steps towards the forward modeling of the thumb joint, 8th World Congress of Biomechanics, Dublin, Ireland (poster presentation, Jul 2018).
- Electromyography (EMG): technique, applications in Sports and Exercise Sciences, Shad Valley Workshop, Human Performance Laboratory, University of Calgary, Canada (Jul 2018).
- Osteoarthritis of the thumb joint: occurrence mechanism and understanding via medical imaging and computer modeling techniques, Human Performance Laboratory Musculoskeletal Biomechanics Seminar, University of Calgary, Calgary, Canada (Mar 2018).

- Mechanical properties of healthy trapeziometacarpal cartilage using indentation testing and contrast-enhanced computed tomography, 11th Symposium of the Hand and Wrist Biomechanics International, Brisbane, Australia (oral presentation, Jul 2017).
- Comparison between intra-articular contact patterns in the trapeziometacarpal joint of healthy and arthritic subjects, OARSI World Congress on Osteoarthritis, Las Vegas, NV, USA (poster presentation, Apr 2017).
- Validating thumb kinematics using dynamic CT & in vivo analysis of pathological joint function, Annual Meeting of the Nederlandse Anatomenvereniging, Lunteren, Netherlands (poster presentation, Jan 2016).
- In vivo contact stress evaluation in the trapeziometacarpal joint using finite deformation biphasic theory and mathematical modeling, 25th Congress of the International Society of Biomechanics, Glasgow, UK (poster presentation, Jul 2015).
- Contact patterns in the trapeziometacarpal joint during isometric tasks obtained via mathematical modeling, 21st Congress of the European Society of Biomechanics, Prague, Czech Republic (oral presentation, Jul 2015).
- Analysing thumb kinematics using dynamic CT: the native and pathological joint, 21st Congress of the European Society of Biomechanics, Prague, Czech Republic (poster presentation, Jul 2015).
- Mathematical modeling of the trapeziometacarpal joint for in vivo stress distribution analysis, 9th Symposium of the Hand and Wrist Biomechanics International, Milan, Italy (oral presentation, Jun 2015).
- The trapeziometacarpal joint: form and function of the native and diseased joint, Annual Meeting of the Nederlandse Anatomenvereniging, Lunteren, Netherlands (poster presentation, Jan 2015).

MENTORSHIP & TEACHING

- PhD student advisor, University of British Columbia (Jun 2020 - Dec 2022)
- Intern advisor, Tidal Medical (Aug 2020 - Nov 2022)
- Medical student mentor, University of Toronto (Dec 2018 - Jun 2020)
- Master student mentor, University of Calgary (Jan 2018 - Sep 2018)
- Undergraduate student supervisor, University of Calgary (Oct 2017 - Dec 2017)
- Teaching assistant, KU Leuven (Sep 2015 - Dec 2016)
- Project advisor, University of Southern California (Sep 2015 - Dec 2015)
- Student panel representative, National Engineering School of Metz (Sep 2009 - Jun 2012)

SIDE PROJECTS

- Prototyped an Android accessibility app for visually impaired users: voice-controlled photo capture with real-time scene descriptions from a generative AI model to support navigation and daily tasks (2023).
- Created a notebook for running Llama 2 locally for text generation (2023).
- Built a computer vision pipeline using YOLO + transfer learning for object detection and tracking (2021).
- Applied dense optical flow (Gunnar Farneback) to quantify motion from video (2020).
- Implemented face detection and tracking for images/videos using Haar cascades and tracking APIs (2019).
- Developed an SVM model to predict coronary heart disease risk; used SMOTE to address class imbalance and reduced type II error by 84% (2018).
- Built an NLP model (Naive Bayes) to classify news articles; achieved 94% accuracy (2017).

CONTINUOUS LEARNING & PROFESSIONAL DEVELOPMENT

- Governing AI Agents, DeepLearning.AI (2025)
- Building Coding Agents with Tool Execution, DeepLearning.AI (2025)
- Agentic AI, DeepLearning.AI (2025)
- Claude Code: A Highly Agentic Coding Assistant, DeepLearning.AI (2025)
- Retrieval Augmented Generation (RAG), DeepLearning.AI (2025)
- Build Apps with Windsurf's AI Coding Agents, DeepLearning.AI (2025)
- Building Generative AI Applications with Gradio, DeepLearning.AI (2023)
- How Diffusion Models Work, DeepLearning.AI (2023)
- LangChain for LLM Application Development, DeepLearning.AI (2023)
- Building Systems with the ChatGPT API, DeepLearning.AI (2023)
- ChatGPT Prompt Engineering for Developers, DeepLearning.AI (2023)
- Fundamentals of Machine Learning for Healthcare, Stanford University (2022)
- Master the Coding Interview: Big Tech (FAANG) Interviews, Zero To Mastery Academy (2022)

- An Entire MBA in 1 Course, Chris Haroun (2021)
- Probability and Statistics for Business and Data Science, Pierian Data Inc. (2020)
- Python for Financial Analysis and Algorithmic Trading, Pierian Data Inc. (2020)
- 101 Crucial Lessons They Don't Teach You In Business School, Chris Haroun (2019)
- Python 3 Deep Dive - Object Oriented Programming, Fred Baptiste (2018)
- Spark and Python for Big Data with PySpark, Pierian Data Inc. (2018)
- Signal Processing Problems, Solved in MATLAB and in Python, Mike X Cohen (2018)
- Data Science Career Guide - Interview Preparation, Pierian Data Inc. (2017)
- Sequence Modeling, DeepLearning.AI (2017)
- Deep Learning: Advanced Computer Vision, Lazy Programmer Inc. (2017)
- Python for Computer Vision with OpenCV and Deep Learning, Pierian Data Inc. (2017)
- Convolutional Neural Networks in TensorFlow, DeepLearning.AI (2017)
- TensorFlow Specialization, DeepLearning.AI (2017)
- Complete TensorFlow 2 and Keras Deep Learning Bootcamp, Pierian Data Inc. (2017)
- PyTorch for Deep Learning with Python Bootcamp, Pierian Data Inc. (2017)
- Introduction to TensorFlow for AI, Machine Learning, and Deep Learning, DeepLearning.AI (2017)
- Cluster Analysis and Unsupervised Machine Learning in Python, Lazy Programmer Inc. (2017)
- Python for Time Series Data Analysis, Pierian Data Inc. (2016)
- Python for Data Science and Machine Learning Bootcamp, Pierian Data Inc. (2016)
- Python and Flask Bootcamp: Create Websites using Flask, Pierian Data Inc. (2016)
- Git & GitHub - The Complete Git & GitHub Course, Code and Create (2016)
- The Complete SQL Bootcamp, Pierian Data Inc. (2016)
- Complete Python Bootcamp, Pierian Data Inc. (2016)
- Machine Learning, Stanford University (2016)
- OpenSim International Workshop, Stanford University (2015)
- Successful Negotiation: Essential Strategies and Skills, University of Michigan (2015)
- Data Analysis Tools, Wesleyan University (2014)